

MICHAEL Z. LAVENDER

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EDUCATION

Boston University – College of Engineering, Boston, MA Expected Graduation: May 2028
B.S. in Electrical Engineering | GPA: 3.98 | Dean’s List All Semesters
Course Highlights: Digital Design (Verilog); Python, MATLAB, & C++; Electronics; Differential Equations; Linear Algebra; Modern Physics
Fall 2026 Courses: Signals & Systems; Graduate-Level Machine Learning; Electricity & Magnetism; Algorithms & Data Structures

PROJECTS

FPGA Neural Network Inference Accelerator | Verilog, Xilinx Vivado, PyTorch Summer 2026
▪ Designing a neural network inference accelerator in RTL Verilog to deploy on a Xilinx Zynq-7020 SoC (PYNQ-Z2).
▪ Integrating a custom multiply-accumulate (MAC) datapath and control FSM to perform classification directly on FPGA hardware, storing all 101K quantized 8-bit weights in on-chip BRAM.
▪ Training a neural network in PyTorch on the MNIST dataset using quantization-aware training with fixed-point (Q1.7) simulation, maintaining 97.3% test accuracy (compared to 97.9% with FP32).
▪ Planning to add real-time webcam-based digit recognition and extend the design to a convolutional neural network.

FPGA-Based Whack-A-Mole Game | Verilog & Xilinx Vivado Spring 2026
▪ Created and implemented a whack-a-mole game in RTL Verilog, deployed on a Xilinx Artix-7 FPGA (Nexys 4 DDR).
▪ Designed an FSM to manage game timing, control a multiplexed 7-segment display, and track the player’s score.
▪ Implemented 16 parallel edge-detection debouncer circuits using a Verilog generate loop, ensuring reliable switch input registration.
▪ Used a 16-bit LFSR seeded from a user-sampled free-running clock-counter to produce a pseudorandom LED activation sequence.
▪ Simulated and verified the modular design using behavioral Verilog testbenches.

C++ Boids Simulator ([GitHub Link](#)) Spring 2026
▪ Developed a Boids simulator, based on Craig Reynolds’ algorithm, to model and visualize the flocking behavior of animals.
▪ Implemented the behavioral algorithm using separation, cohesion, and alignment rules. Extended the original algorithm by implementing predator and prey boids with pursuit and evasion behaviors.
▪ Utilized C++ object-oriented programming concepts (encapsulation, abstraction, and inheritance) to modularize and integrate behavioral and visual systems.

EXPERIENCE

The Auroral Research Group – Boston University ECE Department, Boston, MA April 2026 – Present
Undergraduate Researcher
▪ Investigating small-scale auroral structures using data from high-cadence narrow-field cameras deployed at Poker Flat, AK.
▪ Automated a batch-processing chain, converting ~40 TB of proprietary-format data to HDF5 while generating videos and keograms.
▪ Applying computer vision techniques, including optical flow, to characterize “black” aurora and map plasma flow in the ionosphere.
▪ Extending analysis to data from other instruments, including incoherent scatter radar arrays and NASA’s THEMIS all-sky imagers.

The Chen Lab – Wyss Institute and BU Biological Design Center, Boston, MA July 2025 – December 2025
Undergraduate Researcher
▪ Contributed to research on how Notch signaling affects cell-cell adhesion and vascular homeostasis.
▪ Fabricated 100+ organ-on-a-chip microfluidic devices each month, ensuring timely supply for all experiments.
▪ Collected data through cDNA synthesis and qPCR to quantify gene expression.
▪ Performed a variety of wet-lab techniques, including cell line maintenance, bacterial transformation, and plasmid preparation.

ME Engineers, New York, NY May 2024 – June 2024
Intern
▪ Calculated optimal distances between pipes for a geothermal system installation.
▪ Prepared technical documents for inclusion in a dataset to train an internal artificial intelligence model.

COMMUNITY ENGAGEMENT

▪ **EMT – Boston University EMS.** Deliver pre-hospital emergency care for BU’s student-run EMS service. (September 2024 – Present)
▪ **EMT – Ossining Volunteer Ambulance Corps.** Served on a volunteer ambulance corps while in high school. (2024)
▪ **Firefighter & Apparatus Sergeant – Millwood Fire Company.** Responded to fire and emergency calls. Maintained equipment readiness of the department’s mini-attack apparatus. (2021 – 2025)

ADDITIONAL

▪ **Technical Skills:** Verilog, Vivado, C++, Python, MATLAB, Git, Oscilloscopes, Function Generators, Digital Multimeters, Soldering
▪ **Certifications:** EMT-B (NREMT, MA & NY), BLS Provider CPR & AED (AHA), and Firefighter 1 (Pro Board & NY)
▪ **Hobbies:** Endurance cycling and reading (classic literature, non-fiction, science)